QMIS ERP Full Panel Test Guide — Hierarchy Order

Regenerate PDF: `python3 reports/_build_full_panel_test_guide_pdf.py`

Updated: **18 October 2026** — Phase 3 Ops + Audit closed. Sheet TOTAL **480 / 0 / 0 / 0** of 480.

Ops collections: `/api/v1/ops/{collection}` · Audit: `/api/v1/campus-audit/{collection}` · Alembic **0014**.

Stack: PostgreSQL **18 @ 5433** · API `:8000` · Vite `:5173`.

Module-by-module flow (every sidebar leaf): see

[`QMIS-ERP-Module-Flow-Test-Guide.md`](./QMIS-ERP-Module-Flow-Test-Guide.md)

and PDF `reports/QMIS-ERP-Module-Flow-Test-Guide.pdf`

(each module: **Do → Stored in → Reflects to → Pass**).

Before you start

1. Start Postgres 18 (pgAdmin server on port **5433**). Confirm DB `qmis_erp` exists.

2. Backend:

```
cd backend
./run-local.sh
```

Expect Swagger at http://127.0.0.1:8000/docs and `DATABASE_URL=...@localhost:5433/qmis_erp`.

3. Frontend:

```
cd .. # repo root
npm run dev
```

Open http://localhost:5173

4. Flags (from `.env` / defaults with auth on): `VITE_USE_API_AUTH`, `ADMISSIONS`, `FINANCE`, academics / ops / campus-audit collections (defaults on with auth). Optional: `VITE_USE_API_OPS`, `VITE_USE_API_CAMPUS_AUDIT`.

5. Fresh campus: first API boot seeds roles/demo users. Account Head may need **Settings → Bootstrap campus fee ledger** once.

6. Confirm Alembic head includes **0014_ops_and_audit_records** (`alembic upgrade head`).

Seed logins (development)

Role	Email	Password
Admin	admin@qmis.edu	admin123
HR	hr@qmis.edu	hr12345
Account Head	accounthead@qmis.edu	accounts123
PRM	prm@qmis.edu	prm12345
Director	director@qmis.edu	director123
Principal	principal@qmis.edu	principal123
Coordinator	coordinator@qmis.edu	coord123
Teacher	teacher@qmis.edu	teacher123
Parent	(created on Admission enroll)	(set at enroll / seed)
Student	(optional student user link)	(if provisioned)

Passwords follow `backend/.env` `*_SEED_PASSWORD` if you changed them.

Hierarchy walkthrough (full campus E2E)

Run in this order. **Pass** = API 200, row survives browser refresh, and the next role can see the effect.

Step 1 — Administrator

1. Sign in as **admin@qmis.edu**.
2. **Dashboard** — live KPIs from `/admin/dashboard` (or campus dashboard API).
3. **Class Details** — create/edit a class + section (masters API).
4. **User Creation** — confirm Teacher / Coordinator / Principal logins exist (or create PRM if missing).
5. **Documents / Announcements / Calendar / Communication** — open list (campus APIs).
6. **TC Request Approval** — list hydrates (may be empty).

Step 2 — HR

1. Sign in as **hr@qmis.edu**.
2. **Dashboard** — employee / leave / recruitment KPIs from HR API snapshot (not static filler).
3. **Employees** — open portal list; ensure a teacher employee is linked (or create).
4. **Announcements / Notifications** — lists from HR collections; mark one notification read.
5. **Leave → Received** (`/hr/leave-request/received`) — hydrate via `/leave-requests`; no `schoolerp-leave-requests` in localStorage after load.

Step 3 — Account Head

1. Sign in as **accounthead@qmis.edu**.
2. **Settings → Bootstrap campus fee ledger** (if installments empty).
3. **Fee Collection** — collect one tuition payment; confirm receipt + dashboard KPI updates.
4. **Approvals** — pending counts from finance snapshot.

Step 4 — PRM (Front Office)

1. Sign in as **prm@qmis.edu** (or Admin admissions screens).
2. **Admission Enquiry → convert → Admission → Enroll** (creates student + parent login).
3. Note parent email/password from enroll response / seed.
4. Optional: raise **TC** / gate pass (front-office collections).

Step 5 — Principal (academics SoT — newly Completed)

1. Sign in as **principal@qmis.edu**.
2. **Teacher Allocation** — allocate a teacher to a class/subject; refresh — row still present (`PUT /academics/teacher-allocations`).
3. **Class Timetable** — create timetable with timelines; refresh — still present (`PUT /academics/class-timetables`).
4. **Student Allocation** / Exam Timetable — smoke open (already Completed on sheet).
5. Decide any **timetable change request** if Teacher filed one later.

Step 6 — Coordinator

1. Sign in as **coordinator@qmis.edu**.
2. **Period Chart** / **Lesson Plan Approval** — approve a submitted plan if present.
3. **Class Attendance** — open band/list (attendance API).
4. **Biometric** — Sync demo punches; refresh — rows still present (`PUT /ops/biometric-attendance`). Override remains Teacher-only.

Step 7 — Teacher

1. Sign in as **teacher@qmis.edu**.
2. **Class Attendance** — mark Present/Absent → Save (`POST /attendance`); refresh persists.
3. **Lesson Plan** — submit monthly plan if due.
4. **Mark Entry** — draft/submit marks batch when exam exists.
5. **Class Timetable / TT change** — file a change request; Principal should see it after refresh.
6. **Biometric** — sync + optional override; refresh persists on ops collection.

Step 8 — Director

1. Sign in as **director@qmis.edu**.
2. Approve pending **lesson plans / exams** in inbox if any.
3. **My Attendance** — linked employee (may be empty punches — OK).
4. Tasks / Escalations / Leave smoke.

Step 9 — Parent / Student

1. Parent login from enroll — select child.
2. Timetable / results / attendance read paths (published data only).
3. Student portal if student user exists.

Step 10 — Shared chrome (any staff role)

1. Announcement list · Calendar · Communication inbox · Notifications bell.
2. Leave (mine) · Escalation · Tasks — create one each if time allows.

Step 11 — Ops verticals (module-wise create → refresh)

Use **Admin** for API smoke if role logins are not seeded. Pattern for every screen: open list → **Add** (or edit) → hard refresh (F5) → row still present.

Role / area	Screens to open	Persist via
Librarian	Book List, Issued Books, Members, Announcement, Communication, Calendar, Notifications, My Attendance	`/ops/library-*` + campus chrome APIs
Gate Keeper	Visitors, Inward/Outward, Handover, Vendors, Incidents, Gate Pass list, My Duty, Notifications, My Attendance	`/ops/gate-*` (+ FO gate passes)
Gate Keeper Manager	Assign Duty, Incidents, Team Attendance, Announcement, Leave received	`/ops/gate-*`
Canteen	Dashboard, Menu, Orders, Requests, Announcement	`/ops/canteen-*`
IT Support	Assets, Tickets, Requests, Announcement	`/ops/it-*`
Stationery	Inventory, Issue/Returns, Requests, Announcement	`/ops/stationery-*`
Housekeeping	Inventory, Duty, Schedules, Lost&Found, RO, SOM, Requests, Tasks	`/ops/hk-*` + shared inventory

Transport Manager	Drivers, Vehicles, Routes, Student Transport, Duty, Maintenance, Expenses, Approvals, **Route Data → Track**	`/ops/transport-*` + ProTrack
Driver	My Duty, Fuel, Maintenance, Student Attendance, Route stops	`/ops/transport-driver-*`

Step 12 — JD + Audit suite (module-wise)

Role	Screens	Persist via
Joint Director / Assistant	Dashboard, Employee overview, Request Approvals, Escalations, Meetings, Assets, Tasks, Announcement	`/ops/jd-*`
JD Audit	Templates, Checklist, Question bank, Scoring/Workflow/Visibility, Publish, Planning, Monitoring, Findings	`/api/v1/campus-audit/*`
Process Auditor	My Audits, Schedule, Execute, Observations, RCA, ATR, Escalations, Reports, Communication	`process-*` audit collections
Quality Auditor	Same shape as Process	`quality-*` audit collections

Step 13 — ProTrack365 live bus GPS (how to test)

Already integrated: backend proxy + Transport / Parent / Student Track Bus UI.

1. Confirm `backend/.env` has `PROTRACK_ADMIN_*` and `PROTRACK_PARENT_*` filled (copied from `.env.example` if needed).

2. **Restart API** (`Ctrl+C` then `./run-local.sh`) — `.env` is read at process start.

3. Quick API check (Swagger http://127.0.0.1:8000/docs or curl after login):

- `GET /api/v1/protrack/status` → `configured: true`, `adminConfigured: true`
- `GET /api/v1/protrack/devices?panel=admin` → list of IMEIs / plate numbers
- `GET /api/v1/protrack/track?imeis=<oneImei>&panel=admin` → lat/lng/speed

4. **UI — Transport Manager** (need a transportmanager login, or Admin if you route there):

- Open **Route Data** → **Track** on a route
- Device dropdown should list ProTrack devices
- Map bus marker should jump to live lat/lng; status shows Online/Offline; **Refresh GPS** works

5. **UI — Parent**: login as Parent → **Transport → Track Bus**

- Uses parent panel account; pick a device; live marker updates ~every 15s

6. If amber banner “credentials are not configured” → `.env` empty or API not restarted.

Step 14 — Biometric (what is done vs still open)

Piece	Status
Teacher / Coordinator **Biometric Attendance** page	Wired — Sync + override; rows stored in `/ops/biometric-attendance` (PG)

Live **eSSL / RFID device** import	**Not done** — `RFID_IMPORT_ENABLED=false`; sync button is **simulated** (demo punches), not a real device pull

How to test current Biometric:

1. Login `teacher@qmis.edu` → Attendance → **Biometric Attendance**
2. Click **Sync Biometric Data** → rows update; refresh page → still present
3. Override a row (Teacher only) → refresh → override kept
4. Coordinator path is read-oriented (no override)

When the school enables a real eSSL feed, we turn on `RFID_IMPORT_ENABLED` and add the import job — that is a separate integration.

Per-role smoke checklists

Admin

- [ ] Dashboard KPIs live
- [ ] Class Details write
- [ ] User Creation list
- [ ] Campus Announcement / Calendar / Documents

HR

- [ ] Dashboard from HR snapshot
- [ ] Employees list
- [ ] Notifications mark-read
- [ ] Received Leave API hydrate (no localStorage mirror)

Account Head

- [ ] Finance state hydrate
- [ ] Collect payment → receipt
- [ ] Approvals counts

PRM

- [ ] Enquiry → enroll + parent

Principal

- [ ] Teacher Allocation persists after refresh
- [ ] Class Timetable persists after refresh
- [ ] Decide TT change request

Coordinator / Teacher

- [ ] Attendance / lesson / marks paths
- [ ] Biometric sync persists (`/ops/biometric-attendance`)

Director

- [ ] Approvals / My Attendance linked

Parent / Student

- [ ] Child select + read-only academics

Ops / Audit

- [ ] Library books refresh persist
- [ ] Gate Visitors create → refresh
- [ ] Transport vehicles hydrate
- [ ] Canteen menu hydrate
- [ ] JD Audit templates create → refresh
- [ ] Process observations hydrate
- [ ] Quality observations hydrate

ProTrack365

- [ ] `GET /protrack/status` → configured true (after API restart)
- [ ] Transport Route Data → Track shows live lat/lng
- [ ] Parent Track Bus shows devices from parent panel

Biometric (campus SoT)

- [ ] Teacher Sync → refresh persists
- [ ] Teacher override → refresh persists
- [ ] Live eSSL device — **out of scope until school enables**

Known gaps (do not fail the run)

Area	Status
Live RFID / eSSL device import	**Still needed** — UI/API store simulated punches only (`RFID_IMPORT_ENABLED=false`)
ProTrack365	Integrated (UI + backend); needs API restart after `.env` fill
SMTP / WhatsApp / live payment gateway	Stub / demo mode
staff_ratings REST	Director Star Ratings still FE SoT
Some ops list UIs	Thin tables over campus JSON (CRUD depth varies)

Status matrix

`reports/QMIS-ERP-Role-Module-Status.480-complete.17oct.docx`

`reports/QMIS-ERP-Role-Module-Status.docx` · `erp complete documentation/QMIS-ERP-Role-Module-Status.review.docx`

Quick pass criteria

Pass = HTTP 200 from campus API, data survives hard refresh, and the next role in the hierarchy can observe the change. Biometric and Phase 3 Ops/Audit failures are out of scope for this guide.